

CEN429

Introduction to Data Science

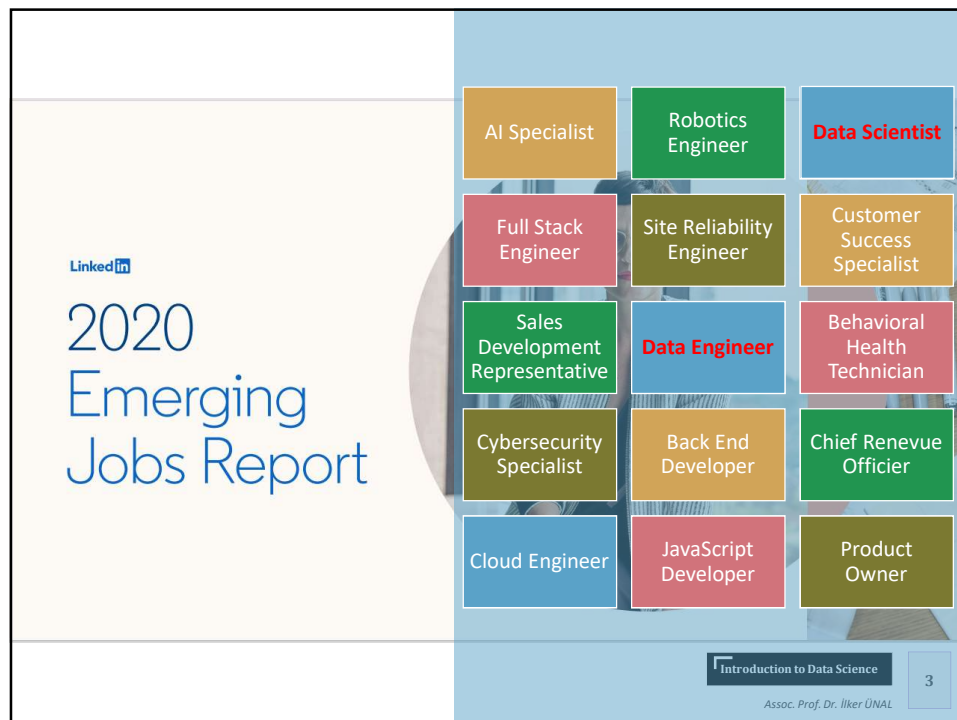
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Five Ws : What, Why, Where, When, Who?



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What is Data? What is Data Science?

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VISUALIZATION
SYST

Data are a set of values of qualitative or quantitative variables about one or more persons or objects.

Data science is an inter-disciplinary field that uses scientific methods, processes, algorithms and systems to extract knowledge and insights from many structured and/or unstructured data.

It is also a "concept to unify statistics, data analysis and their related methods" in order to "understand and analyze actual phenomena" with data

The key word in "Data Science" is not Data, it is Science

Jeff Leek 2013/12/12

One of my colleagues was just at a conference where they saw a presentation about using data to solve a problem where data had previously not been abundant. The speaker claimed the data were "big data" and a question from the audience was: "Well, that isn't really big data is it, it is only X Gigabytes".

While that exact question would elicit groans from most people who work with data, I think it highlights one of the key problems with the thinking around data science. Most people hyping data science have focused on the first word: data. They care about volume and velocity and whatever other buzzwords describe data that is too big for you to analyze in Excel. This hype about the size (relative or absolute) of the data being collected fed into the second category of hype - hype about tools. People threw around EC2, Hadoop, Pig, and had huge debates about Python versus R.

But the key word in data science is not "data", it is "science". Data science is only useful when the data are used to answer a question. That is the science part of the equation. The problem with this view of data science is that it is much harder than the view that focuses on data size or tools. It is much, much easier to calculate the size of a data set and say "My data are bigger than yours" or to say, "I can code in Hadoop, can you?" than to say, "I have this really hard question, can I answer it with my data?"

A few reasons it is harder to focus on the science than the data/tools are:

1. John Tukey's quote: "The combination of some data and an aching desire for an answer does not ensure that a reasonable answer can be extracted from a given body of data.". You may have 100 Gb and only 3 Kb are useful for answering the real question you care about.
2. When you start with the question you often discover that you need to collect new data or design an experiment to confirm you are getting the right answer.
3. It is easy to discover "structure" or "networks" in a data set. There will always be correlations for a thousand reasons if you collect enough data. Understanding whether these correlations matter for specific, interesting questions is much harder.
4. Often the structure you found on the first pass is due to a phenomena (measurement error, artifacts, data processing) that doesn't answer an interesting question.

The issue is that the hype around big data/data science will flame out (it already is) if data science is only about "data" and not about "science". The long term impact of data science will be measured by the scientific questions we can answer with the data.

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BIG DATA

BIG DATA

BIG DATA

- Big data are large size data sets, to manage with these data sets is beyond the ability of commonly used softwares. The term has been in use since the 1990s (John Mashey popularized the term).
- It covers unstructured, semi-structured and structured data, however the main focus is on unstructured data.
- Size of Big data is a constantly increasing through the years, ranging from a few dozen terabytes to many zettabytes of data.
- Handling big data requires the knowledge of a set of integrated techniques and technologies.

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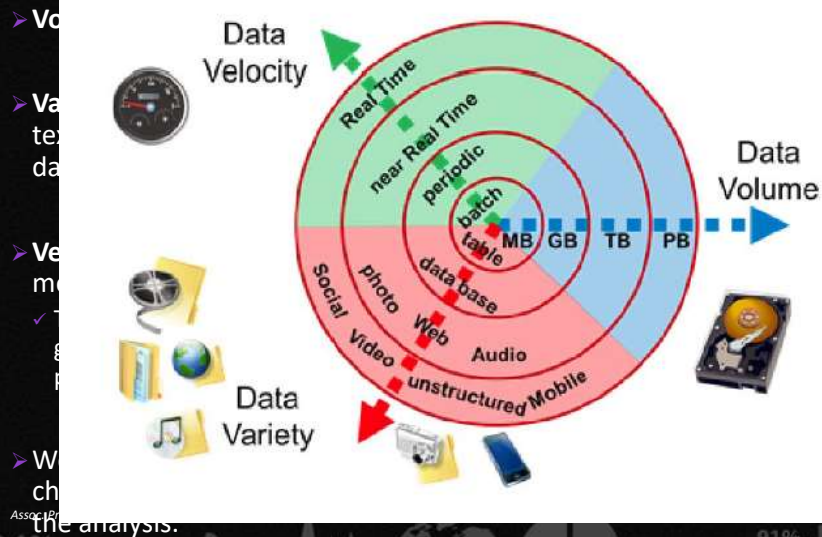
Big data can be described by the following characteristics

- **Volume**, the quantity of generated and stored data.
- **Variety**, the type and nature of the data. In general, Big data draws from text, images, audio, video, etc.. These are the most frequent types of data.
- **Velocity**, the speed at which the data is generated and processed to meet the demands and challenges.
 - ✓ There are two kinds of velocity related to big data: one is the frequency of generation and the other one is the frequency of handling, recording, and publishing.
- We may add another important characteristic, called veracity to these characteristics. It means the quality of data which affects the outcome of the analysis.

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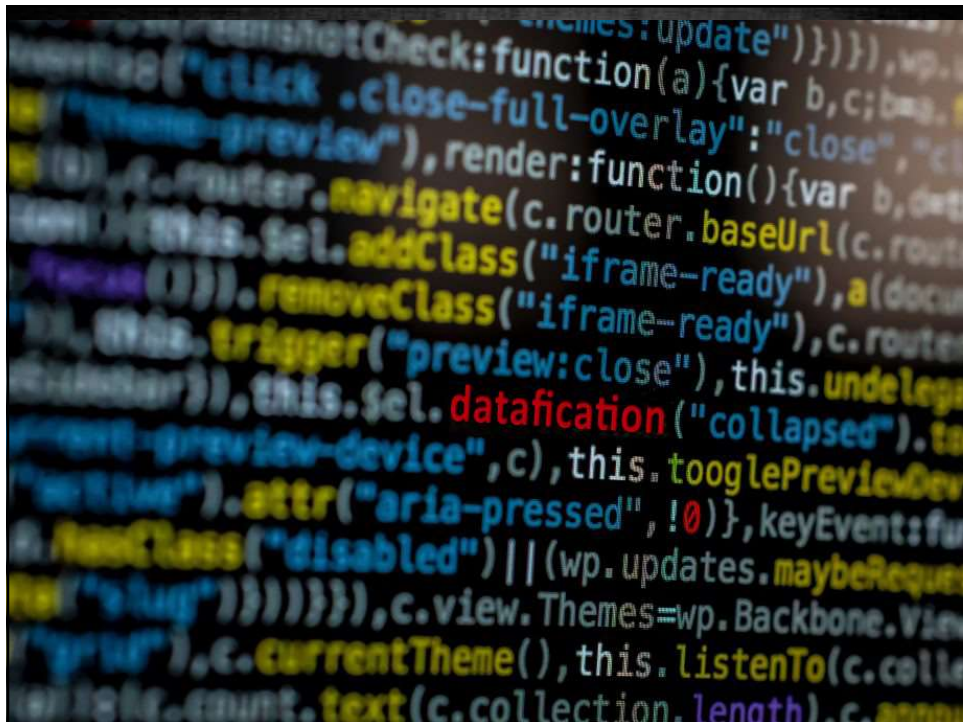
Big data can be described by the following characteristics



Why Is Data Science
Important?

Why Is Data Science Important?

- We have massive amounts of data about many aspects of our lives, the data is being tracked online all the time
 - ✓ shopping, communicating, reading news, listening to music, searching for information, expressing our opinions
- It's not just Internet data,—it's finance, the medical industry, pharmaceuticals, bioinformatics, social welfare, government, education, retail, and the list goes on like that.
- So we need to process these data in order to understand it.
 - ✓ In other words, we need highly skilled data science practitioners.



Datafication is a process of “taking all aspects of life and turning them into data.”

That means when you convert anything to data, you datafy these objects.

For example, Twitter datafies stray thoughts. LinkedIn datafies professional networks

«Once we datafy things, we can transform their purpose and turn the information into new forms of value»

KN Cukier and VM Schoenberger, (2013) «The rise of Big Data» Foreign Affairs

Where do we use Data Science?

- Raw data contain lots of information. Data science is the civil engineering of data
- Here is a short list of common data science uses;

Prediction (predict a value based on inputs)

Classification (e.g., spam or not spam)

Recommendations (e.g., Amazon and Netflix recommendations)

Pattern detection and grouping (e.g., classification without known classes)

Anomaly detection (e.g., fraud detection)

Recognition (image, text, audio, video, facial, ...)

Actionable insights (via dashboards, reports, visualizations, ...)

Automated processes and decision-making (e.g., credit card approval)

Scoring and ranking (e.g., credit score)

Segmentation (e.g., demographic-based marketing)

24% Assoc. P. Optimization (e.g., risk management)

Forecasts (e.g., sales and revenue)

When do we need to use Data Science?

- There are countless situations when you should use Data Science
- 'just because you can, does not mean you should'
 - ✓ If you don't have enough data
 - ✓ If it is a one-off task
 - ✓ If you have too much unlabeled data
 - ✓ If your budget is tight

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Who is Data Scientist?

- The Word of «Scientist» brings science to the mind.
- However, no one calls themselves a data scientist in academia
- A data scientist is someone who knows how to extract meaning from and interpret data, which requires both tools and methods from statistics, programming, machine learning and detailed knowledge about the related domain.
- Since nobody is good at everything, there should be a data science team including different skills (profiles) from different people (areas).

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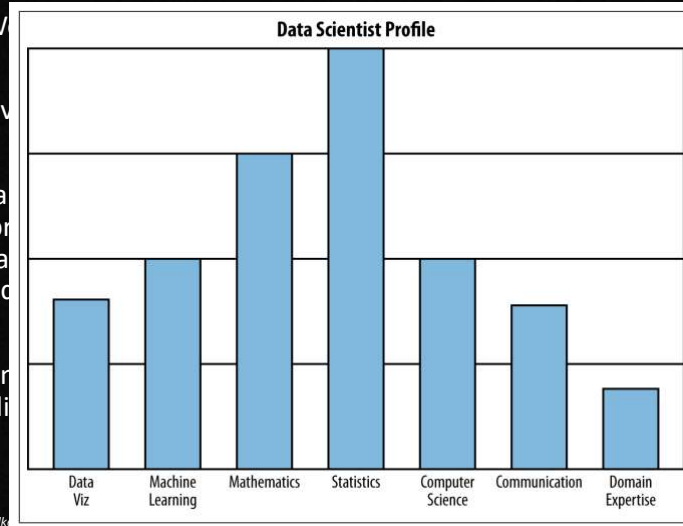
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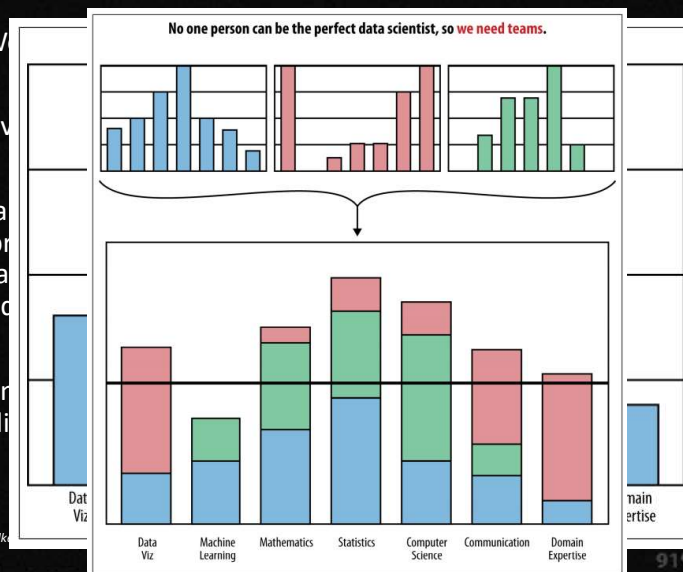
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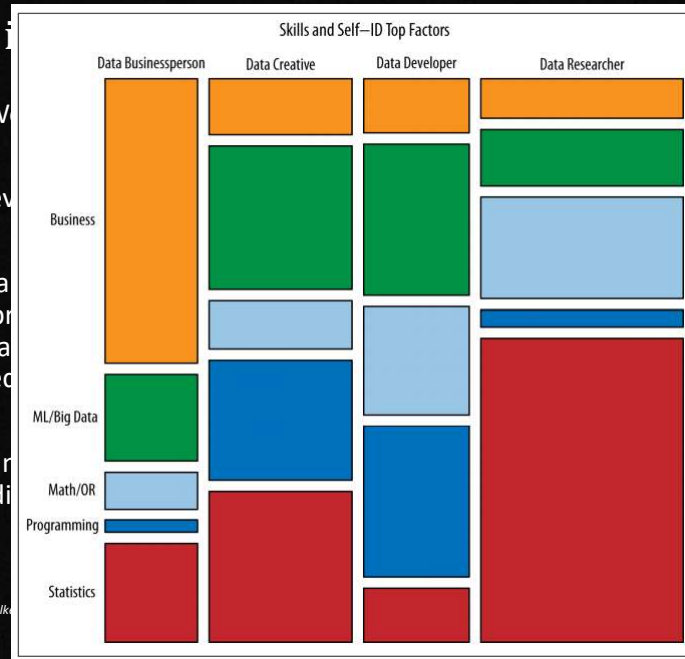
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Data Science Example

David Robinson
Principal Data Scientist at Hadoop, works in R and Python.

- Email
- Twitter
- Github
- Stack Overflow

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Recommended Blogs

- R Bloggers
- RStudio Blog
- R4Stats
- Simply Statistics

Text analysis of Trump's tweets confirms he writes only the (angrier) Android half

I don't normally post about politics (I'm not particularly savvy about polling, which is where data science has had the largest impact on politics). But this weekend I saw a hypothesis about Donald Trump's twitter account that simply begged to be investigated with data:

Todd Vaziri @tvaziri

Every non-hyperbolic tweet is from iPhone (his staff).

Every hyperbolic tweet is from Android (from him).

J. Trump @realDonaldTrump

luck #TeamUSA

ing to New Hampsh

ningCeremony #Rig

about Hillary sayi

twitter.com/mS8qsQT

es

8,3s

1,4

8:59 PM

via Tw

11 AM

via Twi

10:20 PM - Aug 6, 2016

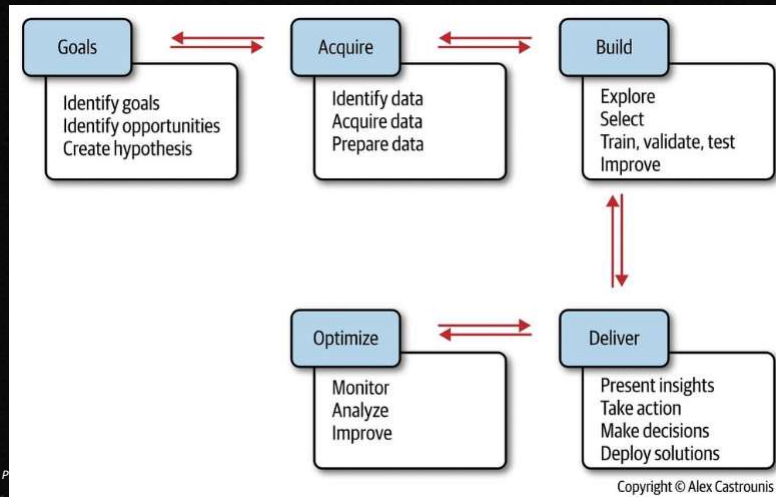
13.3K 9.4K people are Tweeting about this

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The Data Science Process

➤ GABDO models by Alex Castrounis



R

The Comprehensive R Archive Network

Download and Install R

Precompiled binary distributions of the base system and contributed packages, Windows and Mac users most likely want one of these versions of R:

- [Download R for Linux](#)
- [Download R for \(Mac\) OS X](#)
- [Download R for Windows](#)

R is part of many Linux distributions, you should check with your Linux package management system in addition to the link above.

Source Code for all Platforms

Windows and Mac users most likely want to download the precompiled binaries listed in the upper box, not the source code. The sources have to be compiled before you can use them. If you do not know what this means, you probably do not want to do it!

- The latest release (2020-06-22, Taking Off Again) [R-4.0.2 tar.gz](#), read [what's new](#) in the latest version.
- Sources of [R alpha and beta releases](#) (daily snapshots, created only in time periods before a planned release).
- Daily snapshots of current patched and development versions are [available here](#). Please read about [new features](#) and [bug fixes](#) before filing corresponding feature requests or bug reports.
- Source code of older versions of R is [available here](#).
- Contributed extension [packages](#)

Questions About R

- If you have questions about R like how to download and install the software, or what the license terms are, please read our [answers to frequently asked questions](#) before you send an email.

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Why R?

Advantages

- Free of charge, easy to install
- Strong community support
- Up-to-date
- Can complement / can be complemented by other programs
- Requires user knowledge - user thinks about what s/he is doing

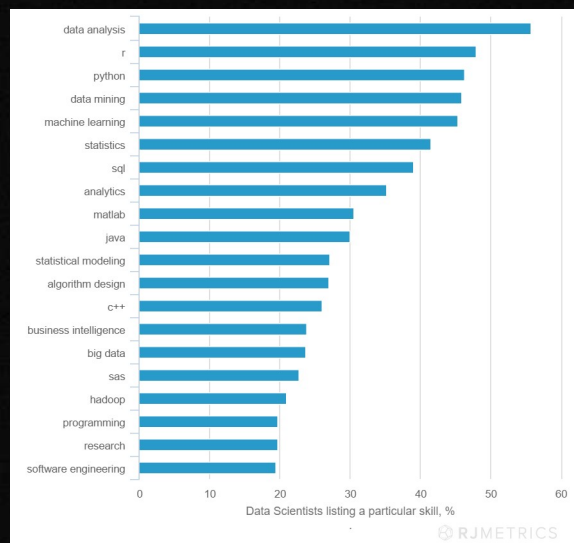
Disadvantages

- Steep learning curve
- Not user friendly
- All objects are stored in the computer memory
- Slower than compiled languages
- Easy to make mistakes, difficult to find the sources of mistakes

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Top 20 Skills of a Data Scientist



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Your Homework

- Download and install R and R studios to your computer
- You can find the installation files in the R CRAN website.

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